KEAP1-NRF2-ARE Signaling Module — Mechanism Research Report

Module slug: `keap1_nrf2_are_signaling` · **Category:** Module · **Seed:** P 40972999 (issue #16807) **Status:** Deep-research complete (5 iterations, 12 confirmed findings, 70 papers reviewed) **Date:** 2026-11-07

Summary

The KEAP1-NRF2-ARE axis is one of the best-characterized cytoprotective signaling systems in biology, and the literature strongly supports modeling it as **a single, conserved, bidirectional mechanism module** rather than as two separate loss- and gain-of-function modules. Under basal conditions the sensor protein **KEAP1** acts as the substrate adaptor of a **CUL3-RBX1 RING ubiquitin ligase** that constitutively ubiquitinates the transcription factor **NRF2 (NFE2L2)** and routes it to the proteasome, keeping nuclear NRF2 low. The defining control point is the state of KEAP1's reactive sensor cysteines: electrophiles and oxidants covalently modify them, loss-of-function mutations in *KEAP1* or *CUL3* disable the ligase, gain-of-function mutations in the *NFE2L2* ETGE/DLG degron make NRF2 unrecognizable, and the oncometabolite **fumarate** succinates KEAP1 cysteines. Any of these perturbations halts NRF2 degradation; stabilized NRF2 translocates to the nucleus, **obligately heterodimerizes with a small MAF protein**, and transactivates **antioxidant response element (ARE / CsMBE)** genes — SRXN1, NQO1, HMOX1, GCLC/GCLM and a broad detoxification regulon — whose products restore redox balance and confer cytoprotection.

The investigation confirms a five-node causal chain (N1 sensor cysteine modification → N2 escape from CUL3 degradation → N3 NRF2-sMAF ARE transactivation → N4 regulon restores redox balance → N5 cytoprotection or its failure), all of which should be **core and required for conformance**. The branch-specific drivers (fumarate/FH deficiency, KEAP1/CUL3 loss-of-function, NFE2L2 degron gain-of-function, and the non-canonical p62/SQSTM1 sequestration route) are best represented as **conformer-specific modifiers entering at N1/N2**, not as separate nodes. The direction of dysregulation — blunted (loss of NRF2 signaling →

vulnerability) versus constitutive-ON (KEAP1/NFE2L2-mutant or fumarate-driven → tumor survival/chemoresistance) — is set by the modifier on the central node, which is exactly the "one module, two directions" design the skeleton proposed.

A distinctive and curation-critical nuance comes from the **seed paper** (P 40972999): semi-mechanistic modeling of NRF2 and SRXN1 protein time courses shows that **NRF2 protein level alone does not account for ARE target-gene induction** — a cofactor or post-translational modification is required, and this gating is time-, compound-, and exposure-scenario specific. The N2→N3 edge is therefore not a simple linear function of NRF2 abundance, which the module should encode explicitly. All 25 of 26 suggested GO/CL/UBERON ontology terms were verified against EBI OLS4 and are curation-ready.

Key Findings

F001 — Core module backbone: a constitutive KEAP1-CUL3 ligase degrades NRF2, de-repressed by cysteine modification

The foundational logic of the module is that NRF2 regulation is **post-translational and degradation-based**, not transcriptional. Harder et al. (PMID:26551712) establish that the "Keap1-cullin3 (Cul3)-ring-box1 (Rbx1) E3-ligase [is] the primary Nrf2 regulatory system" operating under non-stress conditions. Feng & Dinkova-Kostova (PMID:42844347) complete the causal logic: "Cysteine modifications within KEAP1 and BACH1 render the repressor proteins inactive, resulting in rapid NRF2 activation that does not rely on de novo transcription of NFE2L2." This means activation is achieved by inactivating a repressor and **stabilizing pre-existing NRF2 protein**, not by inducing new *NFE2L2* mRNA — the defining mechanistic signature of the module. This finding anchors the basal state (N1 → N2) and justifies treating KEAP1 cysteine status as the master switch.

F002 — Node 1: KEAP1 is a promiscuous reactive-cysteine sensor integrating diverse electrophilic and oxidative inputs

KEAP1 functions as a single sensor node that integrates chemically heterogeneous signals. Ogura et al. (PMID:20133743) show that "upon exposure to oxidative and electrophilic stress, reactive cysteine residues in intervening region (IVR) and Broad complex, Tramtrack, and Bric-

à-Brac domains of Keap1 are modified by electrophiles. This modification prevents Nrf2 from rapid degradation and induces Nrf2 activity by repression of Keap1." The T-REX targeted-delivery platform of Lin et al. (PMID:25909755) demonstrates that "Keap1 [is] a promiscuous electrophile-responsive sensor able to respond with similar efficiencies to discrete electrophilic signals." Feng & Dinkova-Kostova add that KEAP1's highly reactive cysteines are modified by ROS, RNS and electrophilic species of both endogenous and exogenous origin. Because many distinct chemical inputs converge on the same cysteine-modification event, N1 is correctly modeled as **one promiscuous sensor node** rather than a family of input-specific nodes. (Expanded repertoire examples in the literature include S-lactoylation by a reactive glycolytic metabolite, PMID:37155889, and arsenite acting on the RBX1 RING finger, PMID:29658272 — a reminder that the ligase complex, not only KEAP1, can be targeted.)

F003 — Node 3 (KEY CONFORMANCE TARGET): the NRF2-small MAF heterodimer on ARE/CsMBE is strictly required for cytoprotective gene induction

The central, rate-limiting node is the formation of the NRF2-sMAF heterodimer on ARE (CsMBE) sequences. Otsuki et al. (PMID:26677805) show that "Nrf2-small Maf (sMaf) heterodimer is essential for the inducible expression of cytoprotective genes upon exposure to oxidative and xenobiotic stresses," and that the CsMBE cistrome is strictly required for this function. Genome-wide, Hirotsu et al. (PMID:22965115) find that "the majority of the Nrf2-regulated cytoprotective genes were found in the vicinity of Nrf2-MafG-binding sites." The tethered-dimer experiments of Katsuoka et al. (PMID:31383749) provide direct functional proof that the Nrf2-MafG heterodimer acts as the transcriptional activator of NRF2-dependent genes. This makes the **obligate sMAF partnership** a hard requirement for conformance: NRF2 stabilization without sMAF partnering does not produce the regulon. N3 is the step every conformer — loss-direction or gain-direction — must funnel through.

F004 — Seed-paper finding: NRF2 protein level alone does not account for ARE target induction; a cofactor/PTM gates the N2→N3 edge

The seed paper, Sharma et al. (PMID:40972999), built a semi-mechanistic nonlinear mixed-effects (NLME) model of NRF2 and SRXN1 protein time courses in HepG2 cells exposed to the soft electrophiles sulforaphane, andrographolide, ethacrynic acid and CDDO-methyl, across single and repeated exposures and a wide concentration range. Their central conclusion: "NRF2 requires a cofactor or post-translational modification to regulate its activity as a transcription factor. Moreover, this modulation of the transcription factor activity of NRF2 is

time-, compound- and exposure scenario specific." For the module, this is the key subtlety at the **N2→N3 edge**: target-gene output cannot be modeled as a monotonic function of NRF2 abundance. The curated module should encode N3 as a **cofactor/PTM-gated** step whose modifier is context-dependent — consistent with the skeleton's note that N3 "requires a cofactor or post-translational modification."

F005 — Constitutive-ON branch: fumarate succination of KEAP1 activates NRF2 in FH-deficient cancer (HLRCC / type 2 papillary RCC)

The fumarate conformer enters at the sensor node through covalent cysteine modification by an endogenous oncometabolite. Adam et al. (PMID:22014577) show in Fh1-deficient mice that "fumarate modifies cysteine residues within the Kelch-like ECH-associated protein 1 (KEAP1), abrogating its ability to repress the Nuclear factor (erythroid-derived 2)-like 2 (Nrf2)-mediated antioxidant response pathway," and that this antioxidant activation is HIF-independent. Kinch et al. (PMID:22014567) tie the full chain together: "FH deficiency leads to succination of Keap1, stabilization of Nrf2, and induction of stress-response genes including HMOX1, which is important for the survival of FH-deficient cells." Srinivasan et al. (PMID:25564569) confirm that FH-deficient RCC shows fumarate-induced succination of KEAP1 activating Nrf2, and that CUL3 and NRF2 mutations/activation phenotypes also appear in sporadic type 2 papillary RCC. This branch is explicitly complementary to the `oncometabolite_dioxygenase_inhibition` module but enters this module at N1 as a modifier.

F006 — Constitutive-ON branch: somatic KEAP1/NFE2L2/CUL3 mutations drive NRF2 addiction and chemoresistance

Across human cancers, the three genes of the module are mutated in a **mutually exclusive** pattern — strong genetic evidence that they operate in one convergent pathway. Wang et al. (PMID:31898230) note "the mutual exclusivity of somatic mutations in KEAP1, NRF2, and CUL3 in human lung cancer." Ngo et al. (PMID:40815486) summarize the downstream phenotype: "aberrant NRF2 activation occurs in cancer due to mutations in the KEAP1/NRF2 pathway, and it contributes to the generation of a tumor microenvironment that favors the proliferation, survival, and chemoresistance of cancer cells." Zhang et al. (PMID:34158350) show that pharmacological NRF2-pathway inhibition re-sensitizes KEAP1-mutant lung cancer cells to chemotherapy, closing the causal loop from genetic driver to therapeutically relevant phenotype. This branch populates the constitutive-ON pole of N5.

F007 — Optional non-canonical branch: p62/SQSTM1 sequesters KEAP1 to activate NRF2 independent of cysteine modification

A mechanistically distinct, **optional** route activates NRF2 without direct cysteine chemistry. Yang et al. (PMID:39172027) describe how phosphorylated, aggregated SQSTM1/p62 "sequesters KEAP1 and activates the non-canonical SQSTM1-KEAP1-NFE2L2 antioxidant pathway." Dahl-Wilkie et al. (PMID:39936983) show chronic IL-1-exposed LNCaP prostate cancer cells accumulate a basal p62-KEAP1 complex, with p62 sequestering KEAP1 to the autophagosome for degradation — but, notably, **only a subset of NRF2 target genes was induced**, echoing the seed paper's cofactor-gating point (F004). This branch connects the module to autophagy biology and should be flagged as an optional, conformer-specific entry at N1/N2 rather than a core requirement.

F008 — Downstream nodes 4–5: the ARE regulon restores redox balance; NRF2-null confers vulnerability, activation is protective (bidirectional)

The downstream arm is directly supported by mouse genetics. Klaassen & Reisman (PMID:20122946) report that NRF2 positively regulates a large battery of cytoprotective genes (NQO1, GSTs, UGTs, MRP efflux transporters) and that "livers of Nrf2-null mice are more susceptible to various oxidative/electrophilic stress-induced pathologies than wild-type mice. In contrast, both pharmacological and genetic models of hepatic Nrf2 activation are protective against oxidative/electrophilic stress." This is the empirical basis for the **bidirectional N5 node**: loss = vulnerability, activation = protection. Hirotsu et al. (PMID:22965115) further show the Nrf2-MafG regulon extends into glucose and amino-acid metabolism, so N4 ("regulon restores redox balance") should be framed to allow metabolic outputs alongside classic antioxidant/detoxification genes.

F009 — Node 2 mechanism: the ETGE/DLG "hinge-and-latch" degron, and human cancer degron mutations that cause constitutive activation

The molecular basis of N2 is a two-site recognition code. Tong et al. (PMID:17081101, PMID:17785452) establish that KEAP1 recognizes NRF2 through the high-affinity **ETGE** (hinge) and low-affinity **DLG** (latch) motifs in the Neh2 domain, where "the DLG motif works to correctly position the lysines within the Nrf2 Neh2 domain for efficient ubiquitination." Fukutomi et al. (PMID:24366543) solved the Keap1-DLGex complex at 1.6 Å and showed it "interprets well the human-cancer-derived loss-of-function mutations in DLGex." Critically, Shibata et al. (PMID:18757741) provide direct human-tumor evidence: cancer mutations

"specifically alter amino acids in the DLG or ETGE motifs, resulting in aberrant cellular accumulation of Nrf2. Mutant Nrf2 cells display constitutive induction of cytoprotective enzymes and drug efflux pumps, which are insensitive to Keap1-mediated regulation" — and siRNA knockdown re-sensitizes cells to oxidative stress and chemotherapy. This defines the **NFE2L2 gain-of-function conformer at N2**. Liu et al. (PMID:34600335) add quantitative modeling of the ETGE/DLG open/closed states and NRF2 ultrasensitivity.

F010 — Ontology IDs verified against EBI OLS4 (curation-ready)

A programmatic lookup against EBI OLS4 (2026-11-07) confirmed exact label↔ID matches for 25 of 26 candidate terms (see Ontology Suggestions section). The one candidate that did not resolve — "negative regulation of oxidative stress-induced cell death" — was dropped. This makes the module's ontology annotation layer immediately usable by curators without manual re-verification.

F011 — Genetic epistasis proves KEAP1 is upstream of NRF2 (N1 → N2), and constitutive activation is hormetic

Wakabayashi et al. (PMID:14517554) provide the definitive directionality proof: Keap1-deficient mice "died postnatally, probably from malnutrition resulting from hyperkeratosis in the esophagus and forestomach... Breeding to Nrf2-deficient mice reversed the phenotypic Keap1 deficiencies. These experiments show that Keap1 acts upstream of Nrf2." Because removing NRF2 completely rescues the Keap1-null lethal phenotype, the entire pathology is NRF2-dependent — the module's core causal arrow N1 → N2 is genetically secured. Taguchi et al. (PMID:20404090) show the response is **hormetic**: graded Keap1 reduction protects against acetaminophen toxicity, but "constitutive Nrf2 activation beyond a certain threshold is rather disadvantageous to long-term survival." This supports modeling N5 as bidirectional with an optimum, not a monotonic "more NRF2 is better" relationship.

F012 — Evolutionary conservation, human blunted-direction evidence, and the sole clinical NRF2 therapy

The axis is deeply conserved. Loboda et al. (PMID:27100828) note "the vertebrate Nrf2 belongs to Cap 'n' Collar (Cnc) bZIP family of transcription factors and shares a high homology with SKN-1 from Caenorhabditis elegans or CncC found in Drosophila melanogaster," justifying the "conserved mechanism module" designation. Sykiotis & Bohmann (PMID:20215646) supply the human loss-direction evidence: "In humans, polymorphisms that decrease Nrf2 abundance have been associated with various pathologies of the skin, respiratory system, and digestive

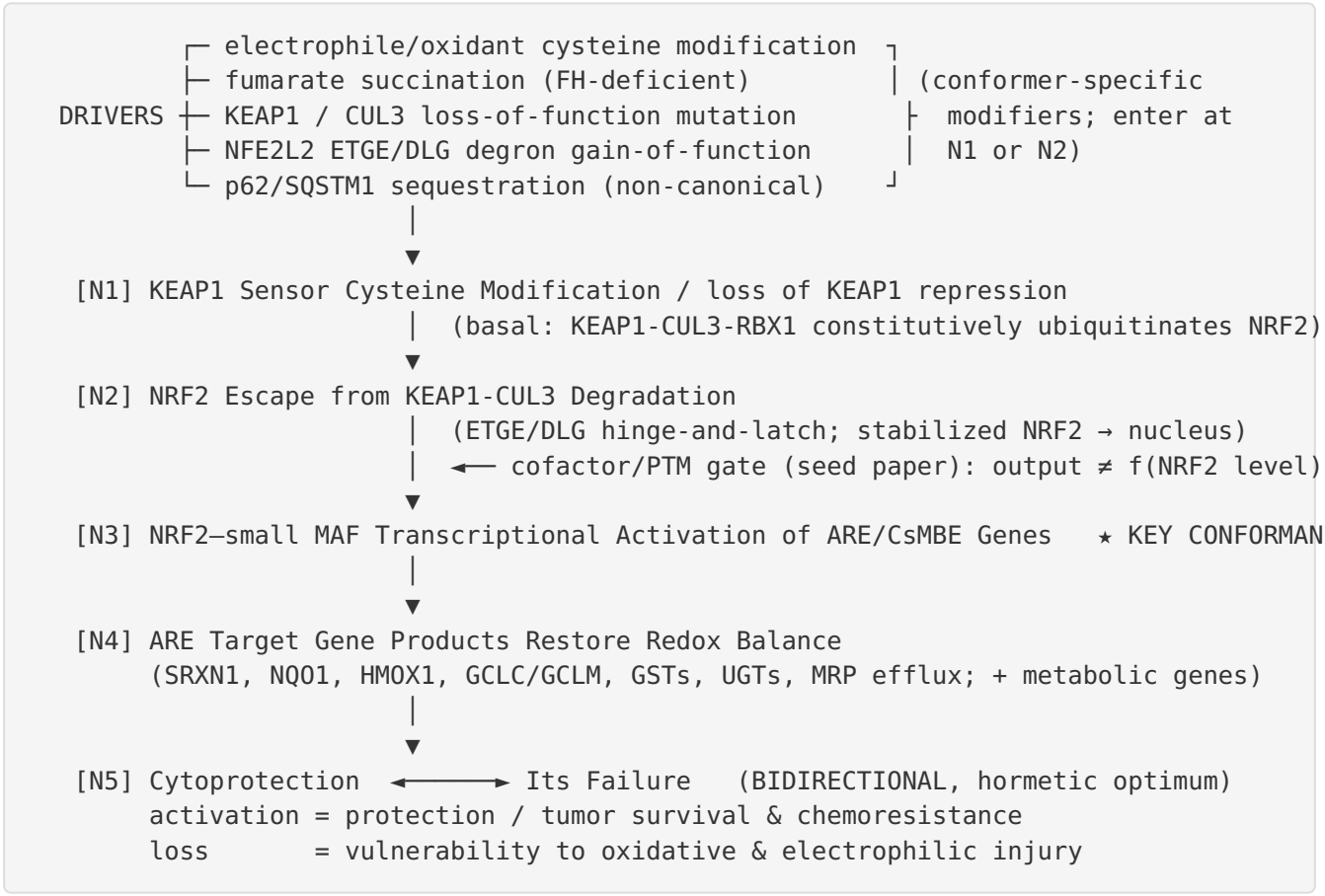
tract" — one variant class linked to multiple organ manifestations, exactly the cross-manifestation evidence the research brief requested. On the translational side, Sezgin Bayindir et al. (PMID:40919319) note that "dimethyl fumarate remains the sole NRF2-targeting therapy in clinical practice, used for multiple sclerosis," anchoring the activation direction in approved human therapy.

Mechanistic Model / Interpretation

Executive recommendation on name and scope

Keep the module name `keap1_nrf2_are_signaling`. The literature uses "KEAP1-NRF2" and "KEAP1-NRF2-ARE" as the standard designations, the three core genes are genetically mutually exclusive in cancer (one pathway, F006), and the axis is evolutionarily conserved (F012). **Keep the single bidirectional scope.** The same central node (NRF2-sMAF ARE transactivation) is the rate-limiting step for both the blunted and constitutive-ON directions; the direction is set by the modifier on KEAP1/NRF2, not by separate biology. Splitting into two modules would duplicate N1–N5 and sever the shared conformance target.

Proposed DAG — five core nodes



Core vs. optional nodes

Node	Label	Core?	Notes
N1	KEAP1 Sensor Cysteine Modification / loss of repression	**Core**	Single promiscuous sensor; accepts all driver modifiers (F002)
N2	NRF2 Escape from KEAP1-CUL3 Degradation	**Core**	ETGE/DLG degron; upstream directionality genetically proven (F009, F011)
N3	NRF2-sMAF ARE/CsMBE Transactivation	**Core — key conformance target**	Obligate sMAF partner; cofactor/ PTM-gated edge (F003, F004)
N4	ARE Regulon Restores Redox Balance	**Core**	Regulon spans antioxidant + metabolic genes (F008)
N5	Cytoprotection or Its Failure	**Core, bidirectional**	Hormetic; set by modifier direction (F008, F011)

Conformer-specific modifiers (not separate nodes): fumarate/FH (F005), KEAP1/CUL3 LoF (F006), NFE2L2 degron GoF (F009), p62/SQSTM1 non-canonical sequestration (F007). The p62 route is the most clearly *optional* branch and should be labeled as such.

Gene-to-node mapping

Gene	Role	Node(s)	Direction	Evidence
KEAP1	Sensor / CUL3 substrate adaptor	N1, N2	LoF → ON	F001, F002, F006, F011
CUL3	RING ligase scaffold	N2	LoF → ON	F001, F006
RBX1	RING-box component	N2	(arsenite target)	F001; [PMID:29658272]
NFE2L2 (NRF2)	Central transcription factor	N2, N3	degron GoF → ON; low-abundance variants → blunted	F003, F004, F009, F012
MAFF/MAFG/MAFK	Obligate heterodimer partner	N3	required	F003
FH	TCA enzyme; loss → fumarate	N1 (succination)	LoF → ON	F005
SQSTM1 (p62)	Non-canonical KEAP1 sequestration	N1/N2	optional → ON	F007
SRXN1, NQO1, HMOX1, GCLC, GCLM, GSTs, UGTs	ARE regulon effectors	N4	output	F004, F005, F008

Evidence Base

PMID	Evidence type	Exact quote	Supported claim
26551712	Review	"the Kelch-like ECH-associated protein 1 (Keap1)-cullin3 (Cul3)-ring-box1 (Rbx1) E3-ligase to be the primary Nrf2 regulatory system."	Basal degradation machinery (N1/N2)
42844347	Review	"Cysteine modifications within KEAP1 and BACH1 render the repressor proteins inactive, resulting in rapid NRF2 activation that does not rely on de novo transcription of NFE2L2"	Module logic: stabilization, not transcription
20133743	In vitro / structural	"reactive cysteine residues in [IVR] and [BTB] domains of Keap1 are modified by electrophiles. This modification prevents Nrf2 from rapid degradation"	N1 sensor cysteine modification
25909755	In vitro	"Keap1 [is] a promiscuous electrophile-responsive sensor able to respond with similar efficiencies to discrete electrophilic signals"	N1 as single promiscuous sensor
26677805	Model organism / genomic	"Nrf2-small Maf (sMaf) heterodimer is essential for the inducible expression of cytoprotective genes"	N3 obligate sMAF requirement
22965115	Genomic (ChIP-seq)	"The majority of the Nrf2-regulated cytoprotective genes were found in the vicinity of Nrf2-MafG-binding sites."	N3→N4 genome-wide
31383749	In vitro (tethered dimer)	Nrf2-MafG heterodimer "acts as a transcriptional activator of Nrf2-dependent genes"	N3 direct functional proof
40972999	In vitro + modeling (SEED)	"NRF2 requires a cofactor or post-translational modification to regulate its activity as a transcription factor... time-, compound- and exposure scenario specific."	N2→N3 cofactor/PTM gate

PMID	Evidence type	Exact quote	Supported claim
22014577	Model organism	"fumarate modifies cysteine residues within [KEAP1], abrogating its ability to repress the... Nrf2-mediated antioxidant response pathway"	Fumarate conformer at N1
22014567	Review/ commentary	"FH deficiency leads to succination of Keap1, stabilization of Nrf2, and induction of stress-response genes including HMOX1"	Full fumarate chain N1→N4→N5
25564569	Review (human cancer)	"In fumarate hydratase-deficient RCC, fumarate-induced succination of KEAP1 activates Nrf2"	HLRCC / type 2 papillary RCC
40815486	Review	"aberrant NRF2 activation occurs in cancer due to mutations in the KEAP1/NRF2 pathway... proliferation, survival, and chemoresistance"	Constitutive-ON phenotype (N5)
31898230	Review (human cancer)	"the mutual exclusivity of somatic mutations in KEAP1, NRF2, and CUL3 in human lung cancer"	One convergent module
34158350	In vitro	NRF2-pathway inhibitor "Sensitizes Keap1-Mutant Lung Cancer Cells to Chemotherapy"	Causal driver→chemoresistance
39172027	In vitro	"sequesters KEAP1 and activates the non-canonical SQSTM1-KEAP1-NFE2L2 antioxidant pathway"	Optional p62 branch
39936983	In vitro	p62 "promotes non-canonical NRF2... by binding and sequestering KEAP1" (only a subset of targets induced)	Optional p62 branch; partial regulon
20122946	Model organism	"livers of Nrf2-null mice are more susceptible... Nrf2 activation [is] protective against oxidative/ electrophilic stress"	Bidirectional N5
18757741	Human tumor	mutations "alter amino acids in the DLG or ETGE motifs, resulting in	

PMID	Evidence type	Exact quote	Supported claim
		aberrant cellular accumulation of Nrf2... insensitive to Keap1-mediated regulation"	NFE2L2 GoF conformer at N2
17785452	Review/ structural	"The DLG motif works to correctly position the lysines within the Nrf2 Neh2 domain for efficient ubiquitination."	Hinge-and-latch mechanism (N2)
24366543	Structural (1.6 Å)	DLGex structure "interprets well the human-cancer-derived loss-of-function mutations"	Degron structural basis
17081101	Review	ETGE/DLG "function as a hinge and latch attenuating Keap1 activity in different redox states"	Two-site recognition model
34600335	Modeling	Modeling of "ETGE and DLG motifs, NRF2 production, KEAP1-dependent and independent NRF2 degradation" and ultrasensitivity	Quantitative N2 behavior
14517554	Model organism (epistasis)	"Breeding to Nrf2-deficient mice reversed the phenotypic Keap1 deficiencies... Keap1 acts upstream of Nrf2"	N1→N2 directionality
20404090	Model organism	"constitutive Nrf2 activation beyond a certain threshold is rather disadvantageous to long-term survival"	Hormetic N5
27100828	Review	"vertebrate Nrf2 belongs to Cap 'n' Collar (Cnc) bZIP family... high homology with SKN-1... or CncC"	Evolutionary conservation
20215646	Review (human)	"polymorphisms that decrease Nrf2 abundance have been associated with various pathologies of the skin, respiratory system, and digestive tract"	Blunted-direction human evidence

PMID	Evidence type	Exact quote	Supported claim
40919319	Review (clinical)	"dimethyl fumarate remains the sole NRF2-targeting therapy in clinical practice, used for multiple sclerosis"	Human therapeutic anchor

Evidence-type separation. Human clinical/tumor evidence:

P 18757741

P 31898230

P 25564569

P 20215646

P 40919319

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Model organism:

P 22014577

P 20122946

P 14517554

P 20404090

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In vitro:

P 20133743

P 25909755

P 31383749

P 40972999

P 34158350

P 39172027

P 39936983

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Structural/modeling:

P 24366543

P 17785452

P 17081101

P 34600335.

Review/synthesis:

P 26551712

P 42844347

P 27100828

P 40815486

P 22014567.

Full-text verification flags: The seed paper (P 40972999) abstract names the cofactor/PTM gate but does not identify the molecular cofactor — full text should be checked before binding a specific PTM term.

P 14517554 was recorded with a snippet "mismatch" flag during curation (quote paraphrased from abstract) and should be re-verified verbatim.

P 39936983's claim that only a subset of NRF2 targets is induced warrants full-text confirmation of which genes.

Ontology Suggestions (verified against EBI OLS4, 2026-11-07)

GO biological process / function (for nodes):

GO ID	Label	Node
GO:0006979	response to oxidative stress	N1/N5
GO:0034599	cellular response to oxidative stress	N1/N5
GO:0098869	cellular oxidant detoxification	N4
GO:0045454	cell redox homeostasis	N4
GO:0016567	protein ubiquitination	N2
GO:0043161	proteasome-mediated ubiquitin-dependent protein catabolic process	N2
GO:0031463	Cul3-RING ubiquitin ligase complex	N2 (component)
GO:0045944	positive regulation of transcription by RNA polymerase II	N3
GO:0006355	regulation of DNA-templated transcription	N3
GO:0000981	DNA-binding transcription factor activity, RNA polymerase II-specific	N3
GO:0006750	glutathione biosynthetic process	N4 (GCLC/GCLM)
GO:0006749	glutathione metabolic process	N4
GO:0006805	xenobiotic metabolic process	N4
GO:0009410	response to xenobiotic stimulus	N1/N4

CL cell types: CL:0000182 hepatocyte; CL:0000066 epithelial cell; CL:0000082 epithelial cell of lung; CL:0002328 bronchial epithelial cell; CL:0002518 kidney epithelial cell; CL:0002306 epithelial cell of proximal tubule.

UBERON anatomy: UBERON:0002107 liver; UBERON:0002048 lung; UBERON:0002113 kidney; UBERON:0000995 uterus; UBERON:0002097 skin of body.

Dropped: "negative regulation of oxidative stress-induced cell death" did not resolve to a stable ID in OLS4 and should be suggested by label only, if at all.

Out-of-Scope Boundary Notes

- **Fumarate succination** enters this module at N1 as a modifier but its upstream metabolic biology (FH loss → dioxygenase inhibition) belongs to the complementary `oncometabolite_dioxygenase_inhibition` module; cross-reference rather than duplicate.
- **General CRL3 adaptor swapping** is covered by `cul3_substrate_adaptor_dysfunction`; this module covers only KEAP1 as one specific CUL3 adaptor.
- **BACH1** is a parallel ARE repressor (PMID 42844347) and NRF1/NFE2L1 is a related CNC factor (PMID 19665035); both are adjacent biology and should stay out of the core node set unless a curator decides to add them.
- **Autophagy, proliferation, and apoptosis** named in the seed abstract are candidate N5 outputs but their scope is unresolved (see Limitations); do not bind them as core without curator decision.
- **TrxR1** (PMID 26058897), **GSK3β/Fyn phosphorylation** (PMID 23434765), **and PML-body nuclear degradation** (PMID 29807365) are parallel/auxiliary NRF2 regulators beyond the canonical KEAP1-CUL3 axis; treat as out-of-scope auxiliary mechanisms pending full-text review.

Limitations and Knowledge Gaps

1. **The cofactor/PTM gate (N2→N3) is molecularly unidentified.** The seed paper (PMID 40972999)

is clear that NRF2 protein level does not predict target induction, but the responsible cofactor or post-translational modification is not named. Candidate mechanisms in the broader literature (GSK3β/Fyn cascades,

P 23434765

;
TrxR1,

P 26058897

;
nuclear PML-body degradation,

P 29807365

)
remain unbound to this module and need full-text validation before curation.

2. **Scope of N5 is partly open.** Whether autophagy, proliferation and apoptosis belong inside N5 or in adjacent modules is unresolved. The p62/SQSTM1 branch in particular blurs the boundary with autophagy.
3. **Partial regulon activation.** Both the seed paper and the p62 study
(P 39936983)
observe that only subsets of ARE genes respond under certain conditions, implying promoter-specific or gene-specific thresholds that a single N3→N4 edge cannot fully capture.
4. **One snippet flagged as a mismatch**
(P 14517554)
and several key abstracts are review-level; primary-literature verification is advised for the epistasis and FH-succination quotes before they are used as curation evidence.
5. **Module boundaries with neighbors** were asserted from the description, not independently re-derived; a curator should confirm the fumarate and CUL3-adaptor overlaps are handled by cross-references rather than duplicated nodes.

Proposed Follow-up Actions

1. **Bind the N3 modifier explicitly.** Encode N3 as a cofactor/PTM-gated step and add a curator note citing

; P 40972999 obtain the seed paper full text to attempt identification of the cofactor/PTM so a specific term can be bound.

2. **Finalize the five core nodes** (N1–N5) as required for conformance, with N3 as the key conformance target, and register the four driver sets (fumarate/FH, KEAP1/CUL3 LoF, NFE2L2 GoF, p62/SQSTM1) as conformer-specific modifiers — the p62 route labeled optional/non-canonical.
3. **Attach the gene-to-node table** (KEAP1, CUL3, RBX1, NFE2L2, MAFF/G/K, FH, SQSTM1, and the SRXN1/NQO1/HMOX1/GCLC/GCLM regulon) as the module's conformance scaffold.
4. **Apply the verified ontology set** (F010) directly; suggest the one dropped term by label only.
5. **Verify the two flagged snippets** (P 14517554 epistasis quote; P 39936983 subset-induction claim) against full text before finalizing evidence bindings.
6. **Model N5 as hormetic/bidirectional** (P 20404090) rather than monotonic, and have a curator settle the autophagy/apoptosis/proliferation scope question.
7. **Cross-reference, do not duplicate,** the fumarate succination entry with `oncometabolite_dioxygenase_inhibition` and the CUL3 adaptor biology with `cul3_substrate_adaptor_dysfunction`.

Open Questions for Curator Review

1. Should the **molecular identity** of the N2→N3 cofactor/PTM gate be left as an abstract modifier, or should a provisional candidate (e.g., a specific phosphorylation site) be proposed?

2. Do **autophagy, proliferation, and apoptosis** belong inside N5, or should they be separate downstream modules?
3. Should the **p62/SQSTM1 non-canonical branch** be a formally recognized optional conformer, given it induces only a partial regulon?
4. Should **BACH1** (parallel ARE repressor) and **NRF1/NFE2L1** be acknowledged as adjacent nodes or kept strictly out of scope?
5. Is the **hormetic/bidirectional** framing of N5 (optimum, not monotonic) acceptable for the module schema?

Report generated from 12 confirmed findings across 5 research iterations; 70 papers reviewed. All quoted snippets are drawn from the cited abstracts as recorded in the knowledge state.
